

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

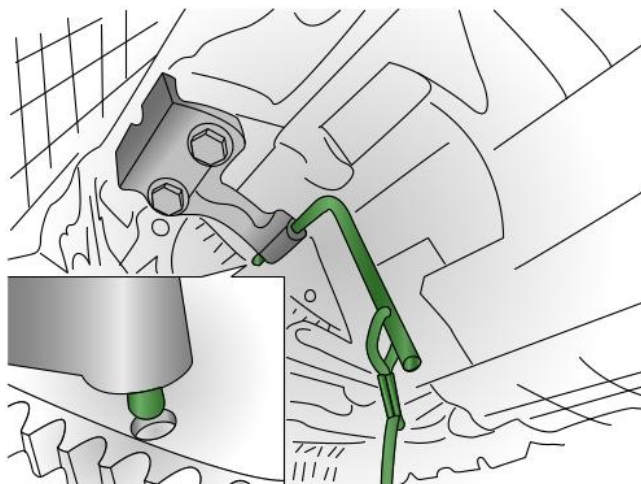
Always use new gaskets

Checking

Turn the crankshaft to TDC for cylinder 1

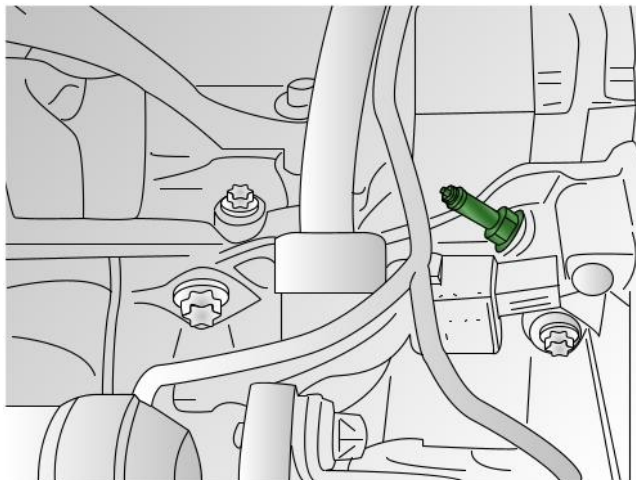
Vehicles with automatic transmission:

Use a special tool to lock the crankshaft (2 365 488, 2 288 380)



Vehicles with manual transmission

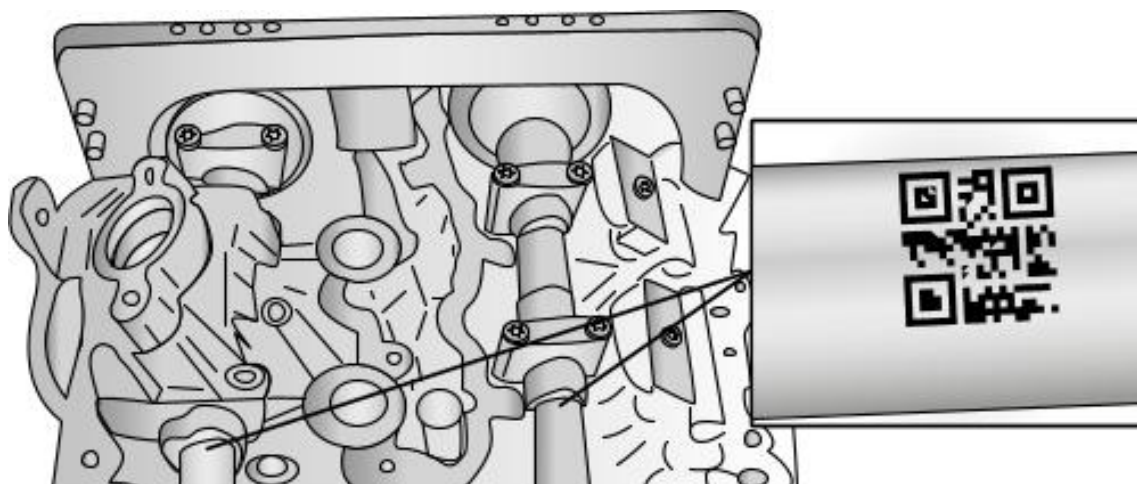
Remove the protective cap



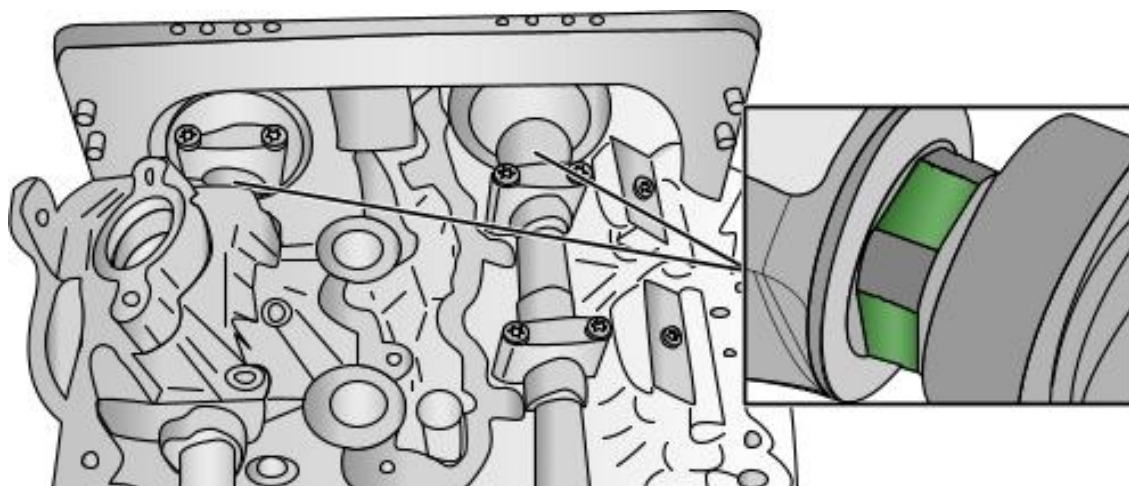
Use a special tool to lock the crankshaft (2 288 380)

All vehicles:

Check the camshaft timing marks

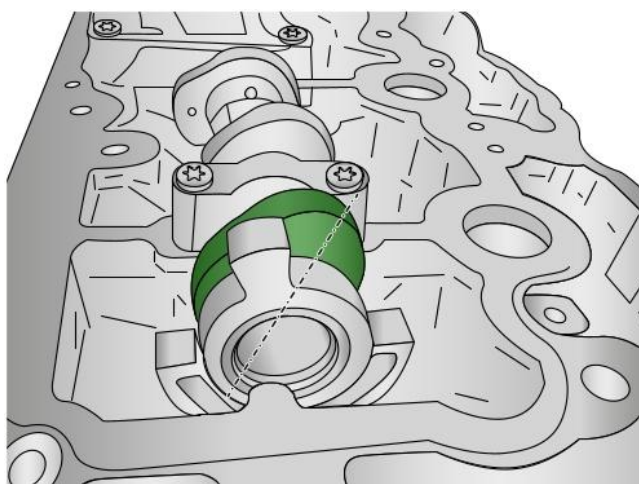
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Align the camshaft flat surfaces vertically

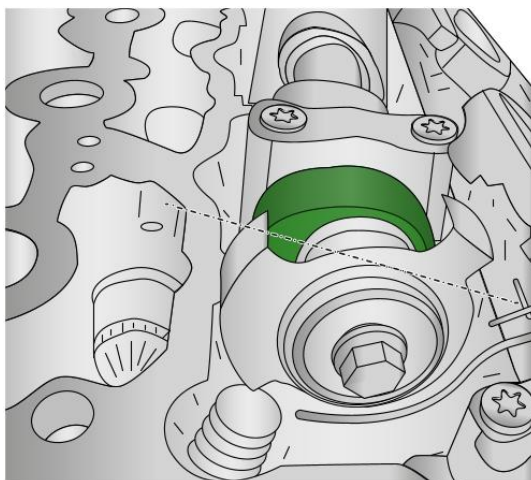


Note: Both camshafts must point upwards at an angle

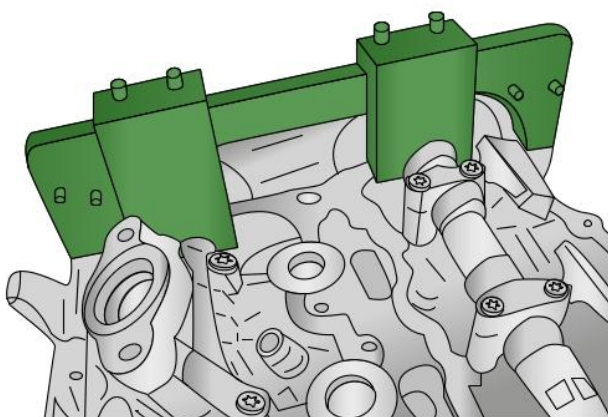
Exhaust camshaft cam angle



Inlet camshaft cam angle

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Fit the camshaft locking tools (2 358 122)



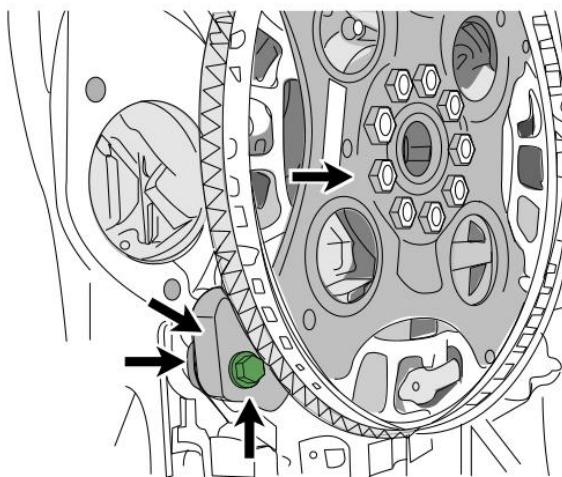
Check that the camshafts are locked

Removal

Remove the engine

Vehicles with automatic transmission:

Fit the locking tool (11 9 260)

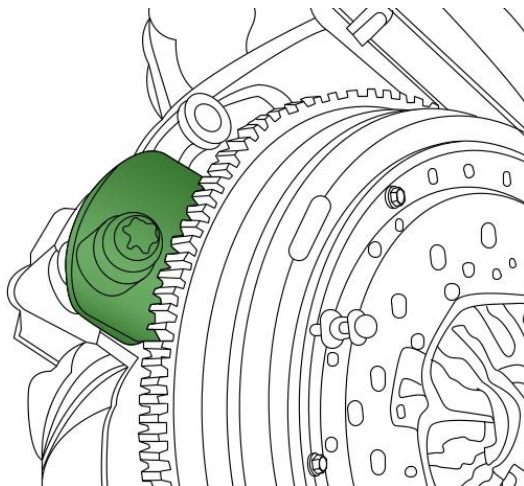


Remove the flexplate

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Vehicles with manual gearbox:

Fit the flywheel locking tool (11 9 810)



Remove the flywheel

All vehicles:

Remove the cylinder head

Remove the oil sump

Remove the oil pump

Vehicles with automatic transmission:

Remove the oil cooler

All vehicles:

Remove the crankshaft position sensor ring

Remove the timing chain cover

Remove the oil pump chain

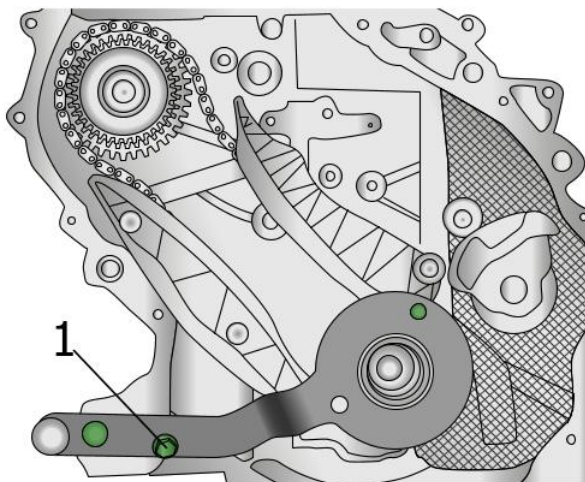
Remove the primary timing chain

Remove the upper timing chain guide

Fit the crankshaft locking tool (2 357 904)

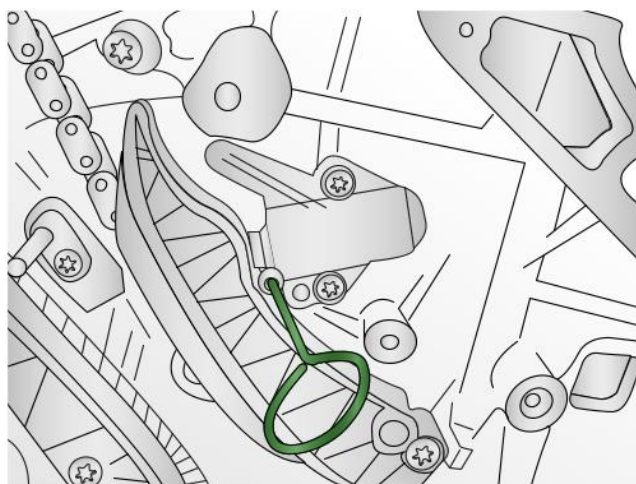
Fit the special tool at the point shown

Fit the bolt (1)



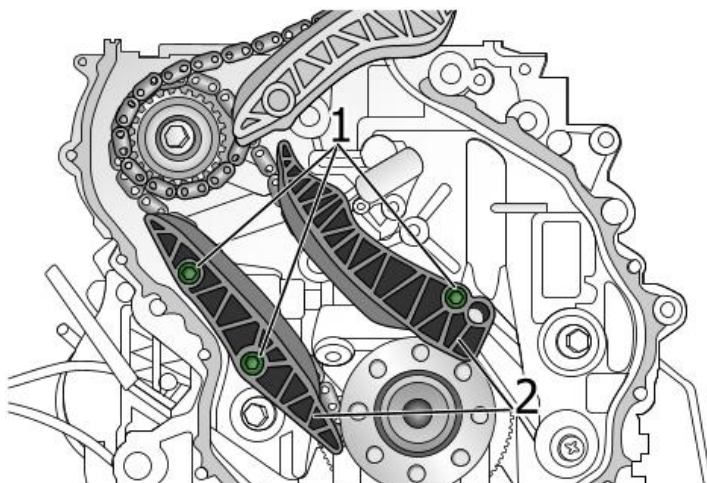
Loosen the chain idler gearwheel bolt

Lock the chain tensioner with a locking pin (11 4 120)

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Remove the secondary chain tensioner

Remove the timing chain guides (1 - 2)



Remove the chain idler gearwheel

Remove the crankshaft locking tool (2 357 904)

Remove the secondary timing chain

Installation

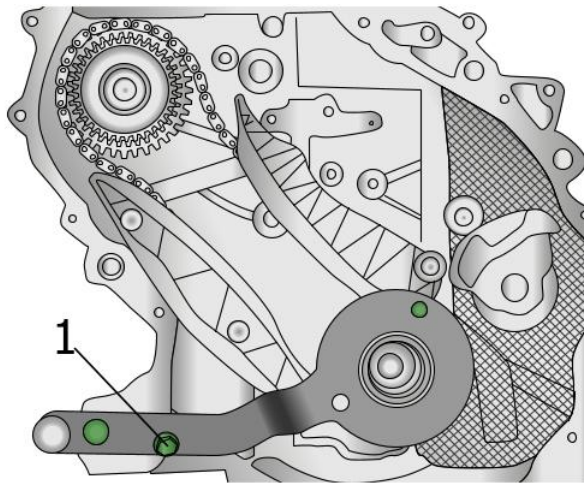
Refit the secondary timing chain

Fit the crankshaft locking tool (2 357 904)

Fit the special tool at the point shown

Fit the bolt (1)

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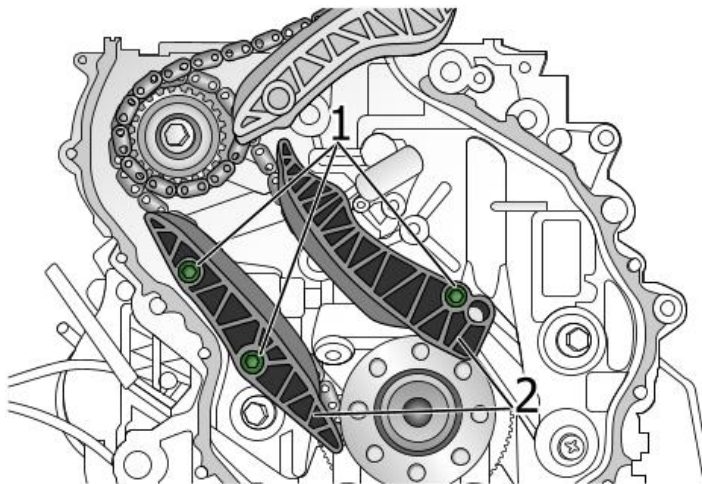


Refit the chain idler gearwheel

Note: Do not tighten the bolt

Refit the chain guides (2)

Renew the guide pin (1)



Refit the secondary timing chain tensioner

Tighten the chain idler gearwheel bolt

Primary timing chain:

Refit the chain guide

Fit the primary timing chain onto the chain idler gearwheel

Refit the oil pump chain

Timing chain cover:

Check the dowel pin(s) for damage

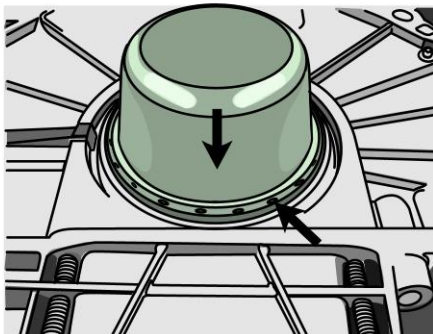
Apply sealant

Refit the timing cover

Crankshaft oil seal:

Use the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Fit the special tool: (11 8 815)

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Note: Do not apply grease

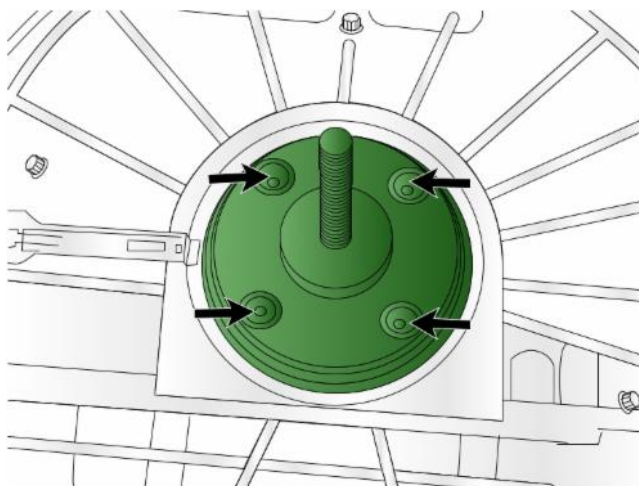
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

Remove the special tool (11 8 815)

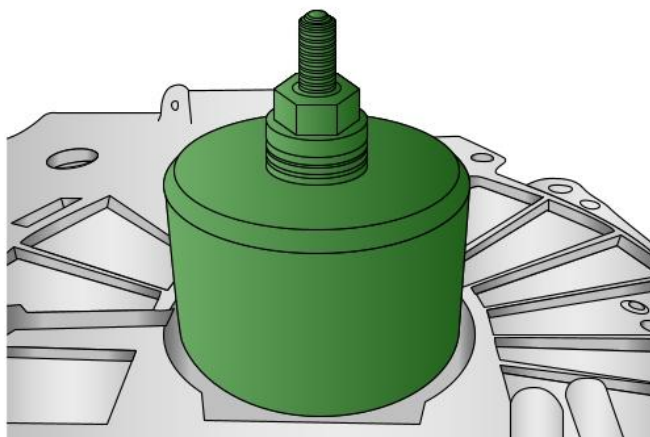
Fit the special tool: (11 8 811)

Tighten the bolts (11 8 812)



Fit the special tool: (11 8 813)

Tighten the nut (11 8 814)



Note: Until the crankshaft seal is level with the timing cover

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Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Further refitting is in the reverse order of removal

Torque settings

Oil sump

Sump 20 (Nm)

Oil pump

Oil pump 19 (Nm)

Oil pump gearwheel

Oil pump gearwheel (Left-hand thread)

Stage 1 5 (Nm)

Stage 2 90 (°)

Timing cover

Timing cover (Rear)

Renew the bolts

Stage 1 8 (Nm)

Stage 2 90 (°)

Timing cover (Front) 10 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper)

Renew the seal

Stage 1 20 (Nm)

Stage 2 25 (°)

Timing chain tensioner (Lower) 10 (Nm)

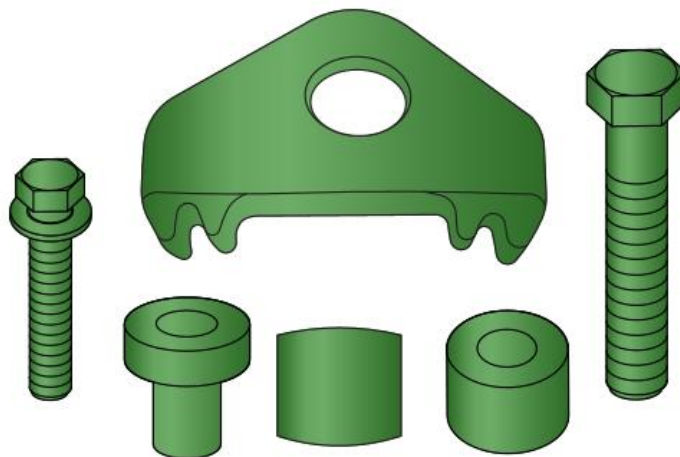
Chain guide rails: (M8: 20Nm)

Use new bolts

Chain idler gearwheel: (M12 x 1.5: 108 Nm)

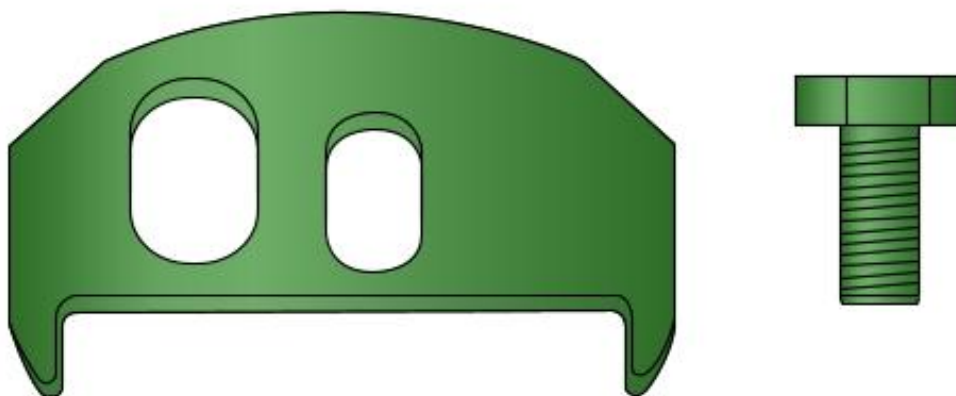
Special tools

Flywheel locking tool: (11 9 260)

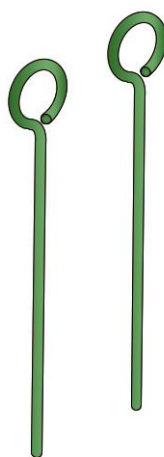


Flywheel locking tool: (11 9 811)

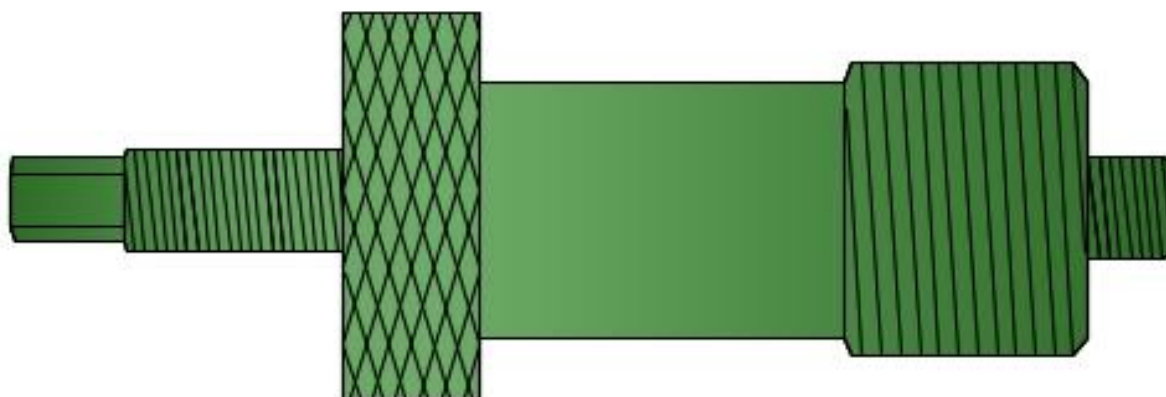
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Locking pin: (11 4 120)

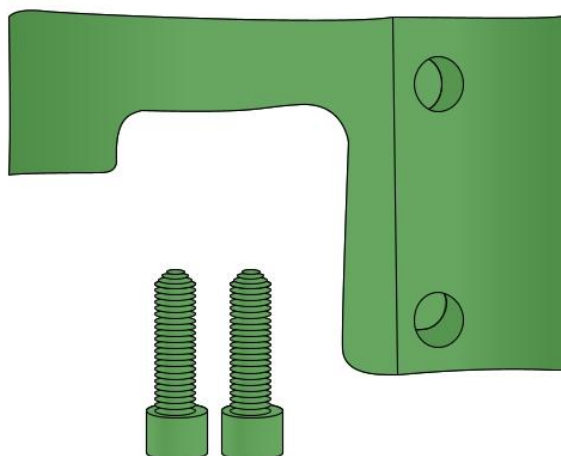


Adaptor: (11 9 340)



Flexplate locking tool: (2 365 488)

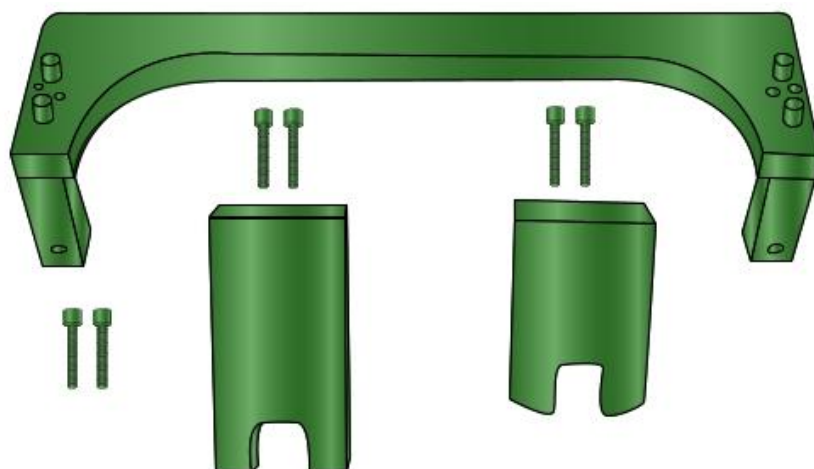
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Flexplate locking tool: (2 288 380)

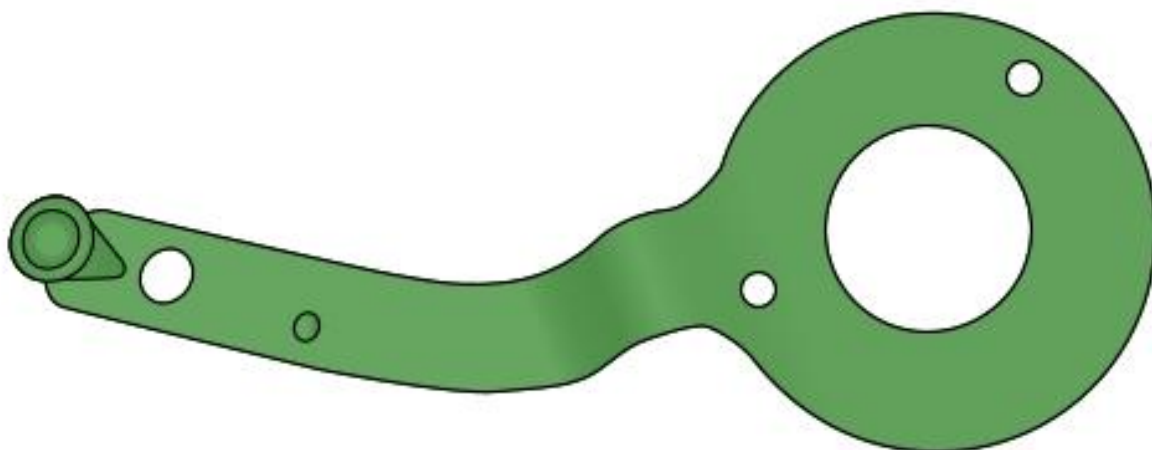


Camshaft blocking tool: (2 358 122)

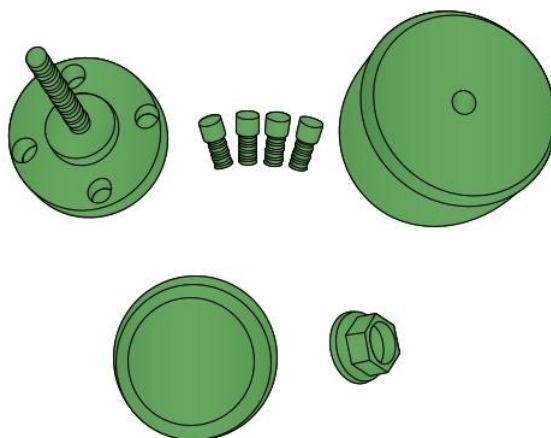


Crankshaft locking tool: (2 357 904)

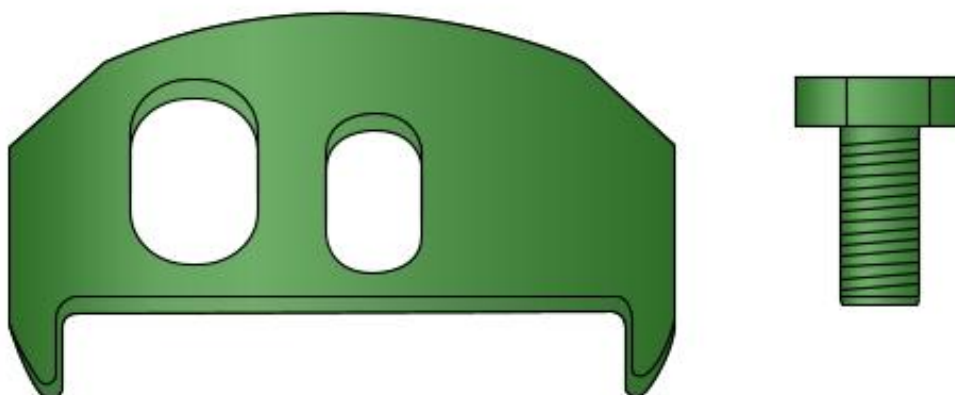
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Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

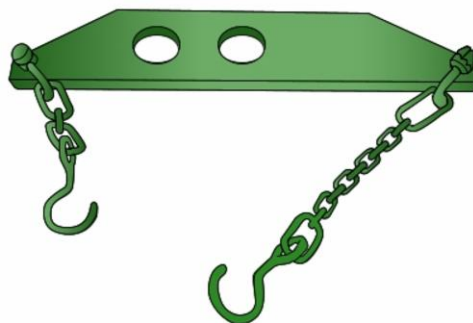


Flywheel locking tool: (11 9 810)

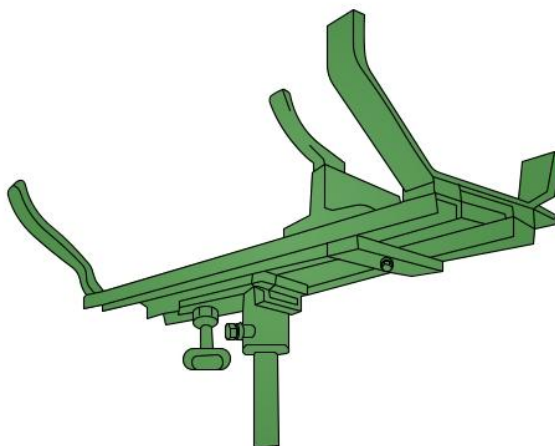


Engine support tool: (11 0 000)

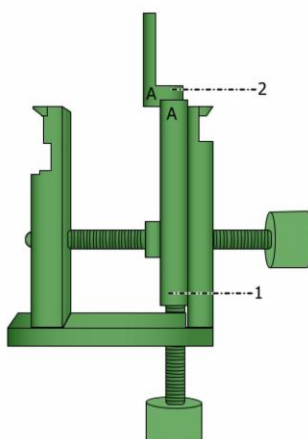
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Gearbox support: (23 4 050, 00 2 030)

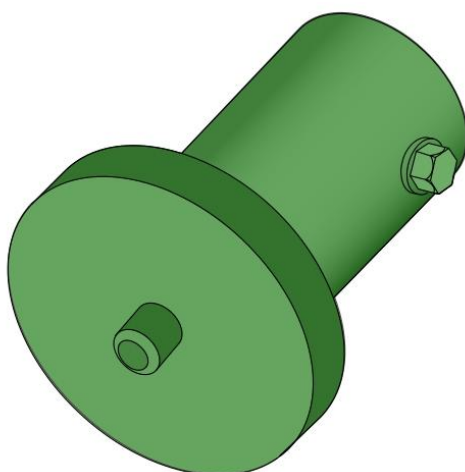


Bearing bracket: (24 4 160, 24 4 161, 24 4 164)

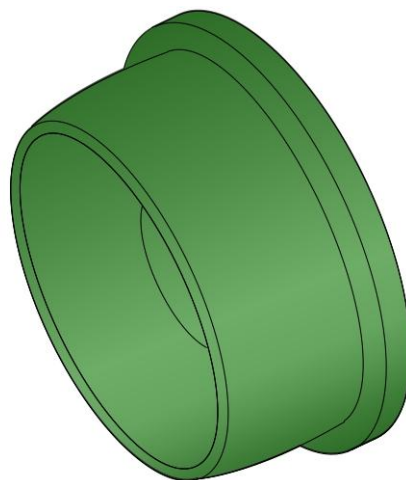


Support (31 2 220)

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Plug (24 2 390)



Plug (24 3 390)

